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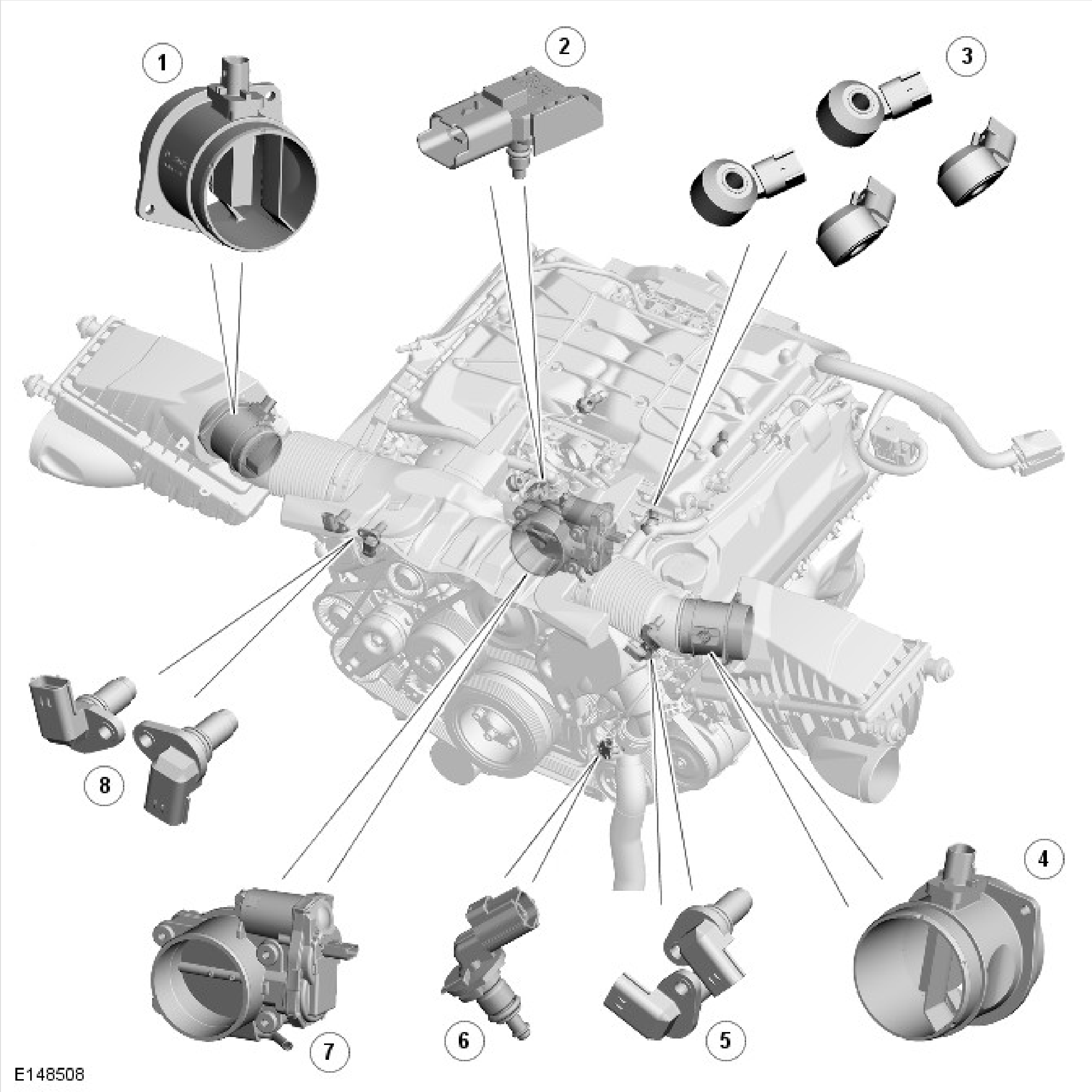
2014.0 RANGE ROVER (LG), 303-14

ELECTRONIC ENGINE CONTROLS - V8 S/C 5.0L PETROL

COMPONENT LOCATION

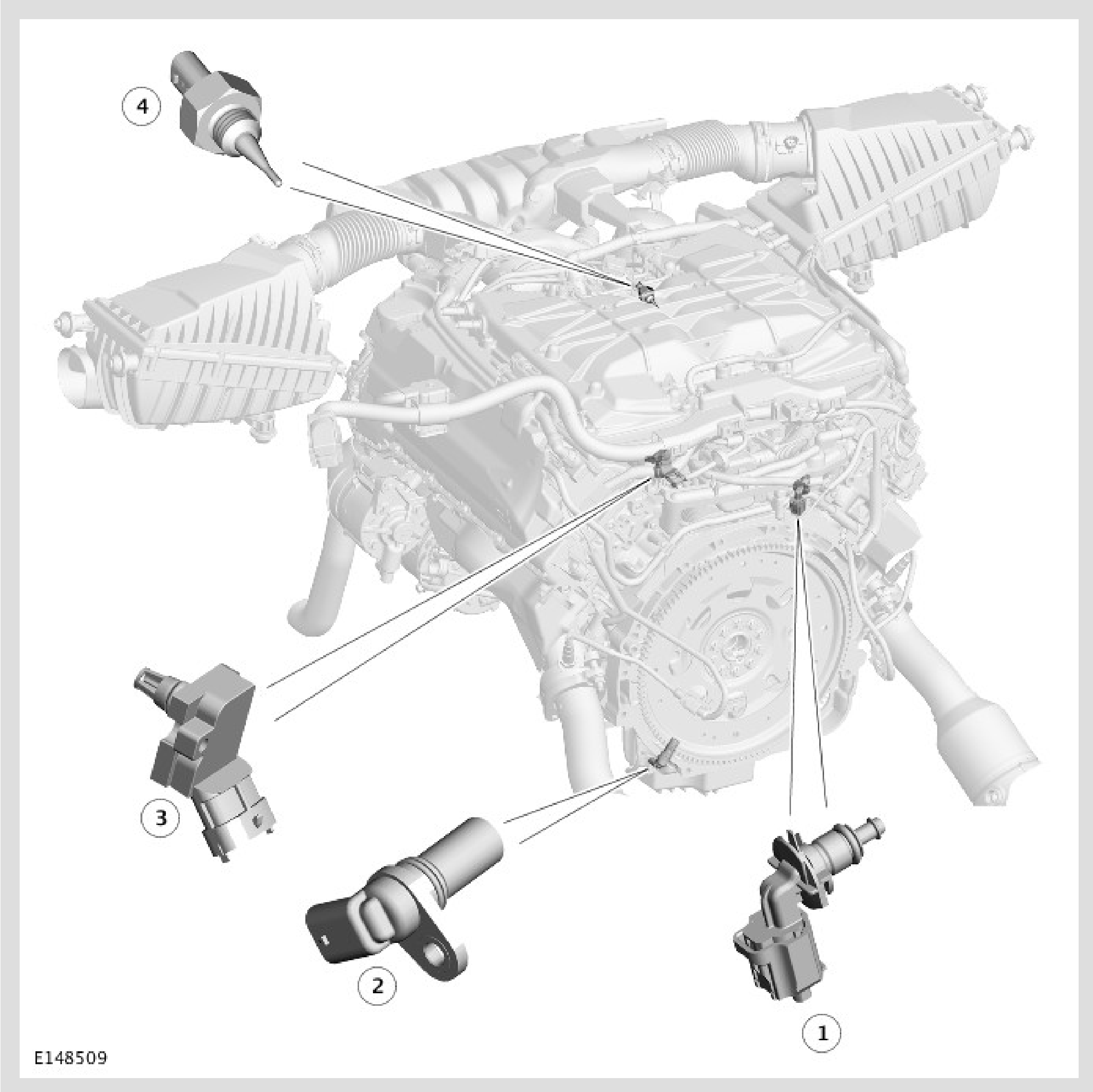
Location 1 of 4



ITEM	DESCRIPTION
1	Right Mass Air Flow and Temperature (MAFT) sensor
2	Manifold Absolute Pressure (MAP) sensor
3	Knock sensor (4 off)
4	Left Mass Air Flow and Temperature (MAFT) sensor
5	Left Camshaft Position (CMP) sensors (2 off)

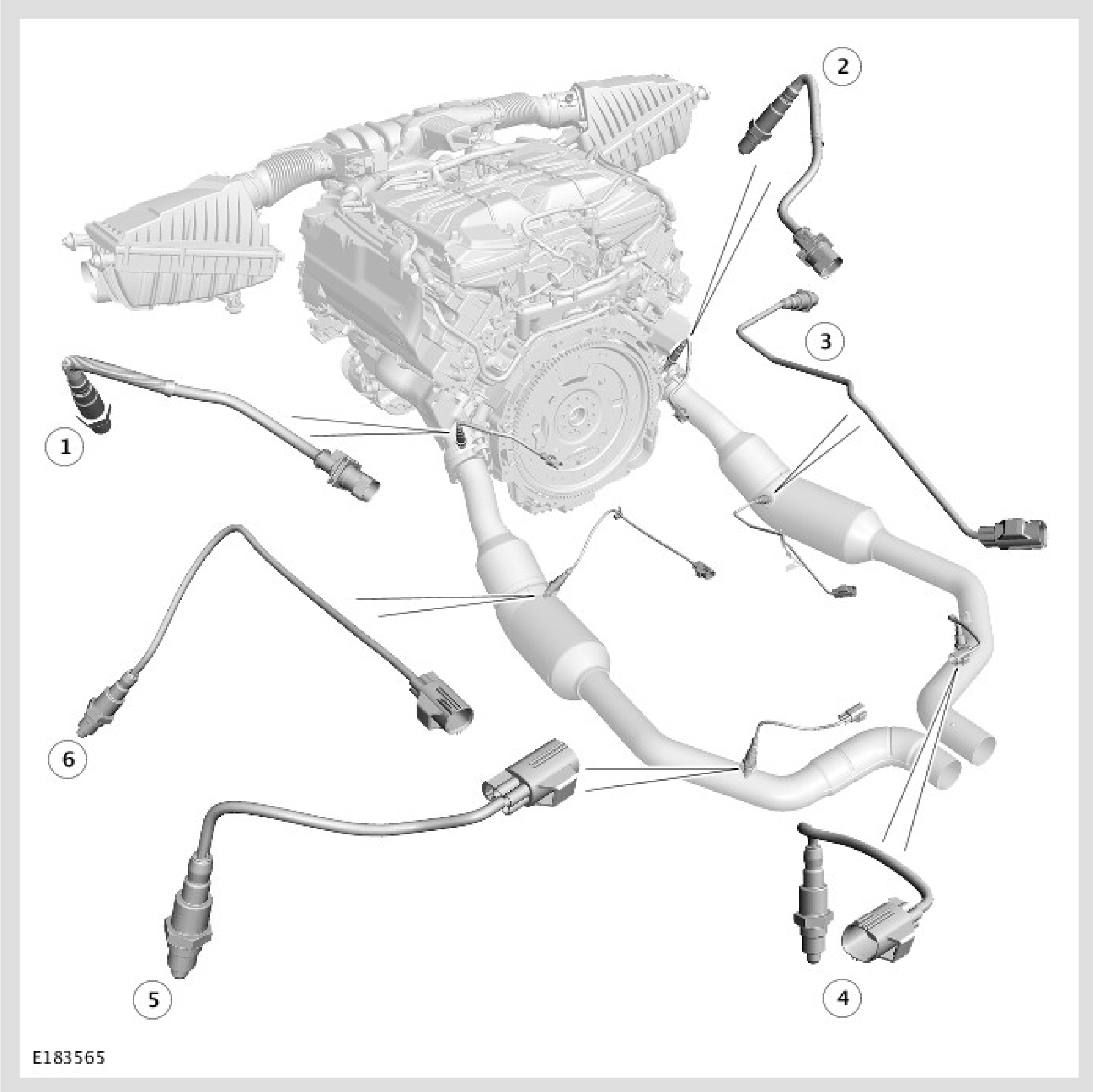
6	Engine Coolant Temperature (ECT) sensor 2
7	Electric throttle
8	Right Camshaft Position (CMP) sensors (2 off)

Location 2 of 4



ITEM	DESCRIPTION
1	Engine Coolant Temperature (ECT) sensor 1
2	Crankshaft Position (CKP) sensor
3	Manifold Absolute Pressure and Temperature (MAPT) sensor
4	Charge air temperature sensor

Location 3 of 4



ITEM	DESCRIPTION
1	Heated Oxygen Sensor (HO2S) - Pre catalyst - Bank 2 sensor 1
2	Heated Oxygen Sensor (HO2S) - Pre catalyst - Bank 1 sensor 1
3	Heated Oxygen Sensor (HO2S) - Mid catalyst - Bank 1 sensor 2
4	Heated Oxygen Sensor (HO2S) - Post catalyst - Bank 1 sensor 3
5	Heated Oxygen Sensor (HO2S) - Post catalyst - Bank 2 sensor 3
6	Heated Oxygen Sensor (HO2S) - Mid catalyst - Bank 2 sensor 2

Location 4 of 4

ITEM	DESCRIPTION
1	Accelerator Pedal Position (APP) sensor
2	Engine Control Module (ECM)
3	Ambient Air Temperature (AAT) sensor

OVERVIEW

The electronic engine control system operates the engine to generate the output demanded by the Accelerator Pedal Position (APP) sensor and loads imposed by other systems.

The electronic engine control system has an Engine Control Module (ECM) that uses a torque-based strategy to evaluate inputs from sensors and other systems, and then produces outputs to engine actuators to produce the required torque.

The electronic engine control system controls the following:

- Charge Air.
- Fueling.
- Ignition Timing.
- Valve Timing.
- Cylinder Knock.
- Idle Speed.
- Engine Cooling Fan.
- Evaporative Emissions.
- On-Board Diagnostic (OBD).
- Immobilization System Interface.
- Speed Control.

DESCRIPTION

ENGINE CONTROL MODULE (ECM)

The Engine Control Module (ECM) is installed in the rear left corner of the engine compartment.

The ECM has the capability of modifying its fuel and ignition control outputs in response to several sensor inputs.

The ECM receives inputs from the following:

- Crankshaft Position (CKP) sensor.
- Camshaft Position (CMP) sensor (4 off).
- Engine Coolant Temperature (ECT) sensor (2 off).
- Knock sensor (4 off).
- Manifold Absolute Pressure (MAP) sensor.
- Manifold Absolute Pressure and Temperature (MAPT) sensor.
- Mass Air Flow and Temperature (MAFT) sensor (2 off).
- Charge air temperature sensor.
- Throttle Position Sensor.
- Heated Oxygen Sensor (HO2S) (6 off).
- Accelerator Pedal Position (APP) sensor.
- Ambient Air Temperature (AAT) sensor.
- Fuel Rail Pressure and Temperature (FRPT) sensor.
  For additional information, refer to: Fuel Charging and Controls (303-04 Fuel Charging and Controls - V8 S/C 5.0L Petrol, Description and Operation).
- Engine cooling fans.
  For additional information, refer to: Engine Cooling (303-03 Engine Cooling - V8 5.0L Petrol/V8 S/C 5.0L Petrol, Description and Operation).
- Brake pedal switch.
  For additional information, refer to: Braking Control System (206-11 Brake Controls, Description and Operation).
- Oil level and temperature sensor.
  For additional information, refer to: Engine (303-01 Engine - V8 S/C 5.0L Petrol, Description and Operation).
- Supercharger bypass valve actuator.
  For additional information, refer to: Intake Air Distribution and Filtering (303-12 Intake Air Distribution and Filtering - V8 S/C 5.0L Petrol, Description and Operation).
- Flex fuel sensor (where fitted).
  For additional information, refer to: Fuel Tank and Lines (310-01 Fuel Tank and Lines - V8 5.0L Petrol/V8 S/C 5.0L Petrol, Description and Operation).
- Low Pressure (LP) fuel sensor.
  For additional information, refer to: Fuel Tank and Lines (310-01 Fuel Tank and Lines - V8 5.0L Petrol/V8 S/C 5.0L Petrol, Description and Operation).
- Fuel Pump Driver Module (FPDM).
  For additional information, refer to: Fuel Tank and Lines (310-01 Fuel Tank and Lines - V8 5.0L Petrol/V8 S/C 5.0L Petrol, Description and Operation).

- Transmission Control Switch (TCS).
  For additional information, refer to: Transmission Description (307-01 Automatic Transmission/Transaxle - Vehicles With: 8HP70 8-Speed Automatic Transmission AWD, Description and Operation).
- Restraints Control Module (RCM).
  For additional information, refer to: Air Bag and Safety Belt Pretensioner Supplemental Restraint System (SRS) (501-20 Supplemental Restraint System, Description and Operation).

The ECM provides outputs to the following:

- Electric throttle.
- ECM relay.
- Fuel injector (8 off).
  For additional information, refer to: Fuel Charging and Controls (303-04 Fuel Charging and Controls - V8 S/C 5.0L Petrol, Description and Operation).
- Ignition coil (8 off).
  For additional information, refer to: Engine Ignition (303-07 Engine Ignition - V8 5.0 L Petrol/V8 S/C 5.0L Petrol, Description and Operation).
- Variable Camshaft Timing (VCT) actuator (4 off).
  For additional information, refer to: Engine (303-01 Engine - V8 S/C 5.0L Petrol, Description and Operation).
- Supercharger bypass valve actuator.
  For additional information, refer to: Intake Air Distribution and Filtering (303-12 Intake Air Distribution and Filtering - V8 S/C 5.0L Petrol, Description and Operation).
- Evaporative Emission (EVAP) purge valve.
  For additional information, refer to: Evaporative Emissions (303-13 Evaporative Emissions - V8 5.0L Petrol/V8 S/C 5.0L Petrol, Description and Operation).
- Engine starter relay.
  For additional information, refer to: Starting System (303-06 Starting System - V8 5.0L Petrol/V8 S/C 5.0L Petrol, Description and Operation).
- Engine cooling fans.
  For additional information, refer to: Engine Cooling (303-03 Engine Cooling - V8 5.0L Petrol/V8 S/C 5.0L Petrol, Description and Operation).
- Charge air coolant pump relay.
  For additional information, refer to: Supercharger Cooling (303-03 Supercharger Cooling - V8 S/C 5.0L Petrol, Description and Operation).
- High Pressure (HP) fuel pump (2 off).
  For additional information, refer to: Fuel Charging and Controls (303-04 Fuel Charging and Controls - V8 S/C 5.0L Petrol, Description and Operation).

- FPDM.
  For additional information, refer to: Fuel Tank and Lines (310-01 Fuel Tank and Lines - V8 5.0L Petrol/V8 S/C 5.0L Petrol, Description and Operation).
- Diagnostic Module Tank Leakage (DMTL) pump (where fitted).
  For additional information, refer to: Evaporative Emissions (303-13 Evaporative Emissions - V8 5.0L Petrol/V8 S/C 5.0L Petrol, Description and Operation).

CRANKSHAFT POSITION (CKP) SENSOR

The Crankshaft Position (CKP) sensor is an inductive sensor that allows the Engine Control Module (ECM) to determine the angular position of the crankshaft and the engine speed.

The CKP sensor is installed in the rear left side of the oil pan, in line with the engine drive plate. The sensor is secured with a single screw and sealed with an O-ring. A three pin electrical connector provides the interface with the engine harness.

The head of the CKP sensor faces a reluctor ring pressed into the outer circumference of the engine drive plate. The reluctor ring has 60 teeth, with two teeth missing. There are 58 teeth at 6° intervals, with two teeth removed to provide a reference point with a centerline that is 21° Before Top Dead Center (BTDC) on cylinder 1 of bank 2.

If the CKP sensor fails, the ECM:

- Uses signals from the Camshaft Position (CMP) sensors to determine the angular position of the crankshaft and the engine speed.
- Adopts a limp home mode where engine speed is limited to a maximum of 3000 Revolutions Per Minute (RPM).

With a failed CKP sensor, engine starts will require a long crank time while the ECM determines the angular position of the crankshaft using the CMP sensors.

CAMSHAFT POSITION (CMP) SENSOR

The Camshaft Position (CMP) sensors are Magneto Resistive Element (MRE) sensors that allow the Engine Control Module (ECM) to determine the angular position of the camshafts. MRE sensors produce a digital output which allows the ECM to detect speeds down to zero.

The four CMP sensors are installed in the front upper timing covers, one for each camshaft.

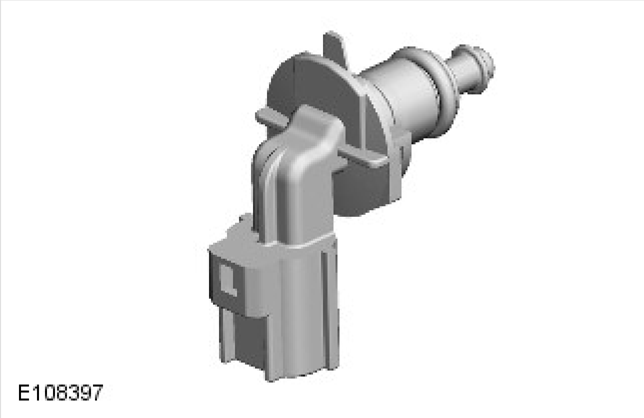
Each CMP sensor is secured with a single screw and sealed with an O-ring. On each CMP sensor, a three pin electrical connector provides the interface with the engine harness.

The head of each CMP sensor faces a reluctor ring attached to the front of the related Variable Camshaft Timing (VCT) actuator.

If an exhaust CMP sensor fails, the ECM disables the VCT of both exhaust camshafts.

If an intake CMP sensor fails, the ECM disables the VCT of both intake camshafts. This can result in reduced engine performance and slow to start. A failed CMP sensor will not prevent the engine starting.

ENGINE COOLANT TEMPERATURE (ECT) SENSOR



The Engine Cooling Temperature (ECT) sensors are Negative Temperature Coefficient (NTC) thermistors that allow the ECM to monitor the engine coolant temperature.

There are two identical ECT sensors installed, which are identified as ECT 1 and ECT 2. Each sensor is secured with a twist-lock and latch mechanism, and is sealed with an O-ring. A two pin electrical connector provides the interface between the sensor and the engine harness.

Engine Coolant Temperature (ECT) Sensor 1

ECT 1 is installed in the heater manifold, at the rear of the bank 1 cylinder head. The input from this sensor is used in calibration tables and by other systems.

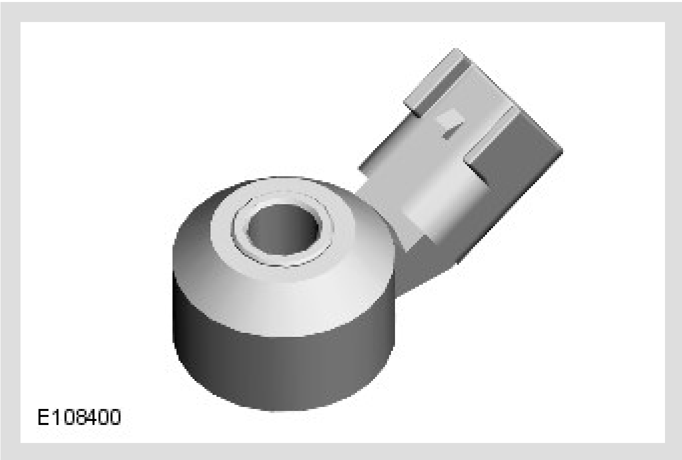
If there is an ECT 1 fault, the ECM adopts an estimated coolant temperature. On the second consecutive trip with an ECT 1 fault, the ECM illuminates the Malfunction Indicator Lamp (MIL).

Engine Coolant Temperature (ECT) Sensor 2

ECT 2 is installed in the lower hose connector which attaches to the bottom of the thermostat. The input from this sensor is used for On-Board Diagnostic (OBD) 2 diagnostics and, in conjunction with the input from ECT 1, to confirm that the thermostat is functional.

If there is an ECT 2 fault, the ECM illuminates the MIL on the second consecutive trip.

KNOCK SENSORS



The knock sensors are piezo-ceramic sensors that allow the Engine Control Module (ECM) to employ active knock control and prevent engine damage from pre-ignition or detonation.

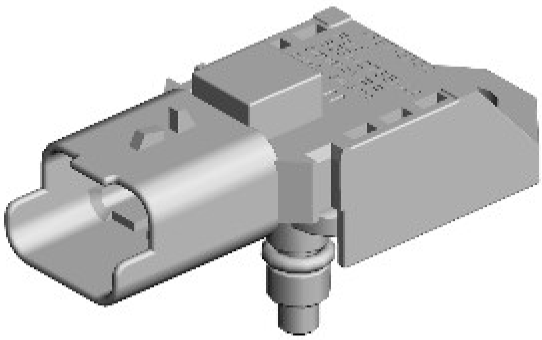
Four knock sensors are installed on the inboard side of each cylinder head, two mid-way between the front two cylinders, and two mid-way between the rear two cylinders. Each knock sensor is secured with a single screw. On each knock sensor, a two pin electrical connector provides the interface with the engine harness.

The ECM compares the signals from the knock sensors with mapped values stored in memory to determine when detonation occurs on individual cylinders. When detonation is detected, the ECM retards the ignition timing on that cylinder for a number of engine cycles, then gradually returns it to the original setting.

The ECM cancels closed loop control of the ignition system if the signal received from a knock sensor becomes implausible. In these circumstances the ECM defaults to base mapping for the ignition timing. This ensures the engine will not become damaged if low quality fuel is used. The Malfunction Indicator Lamp (MIL) will not illuminate, although the driver may notice that the engine 'pinks' in some driving conditions and displays a drop in performance and smoothness.

The ECM calculates the default value if one sensor fails on each bank of cylinders.

MANIFOLD ABSOLUTE PRESSURE (MAP) SENSOR



E148520

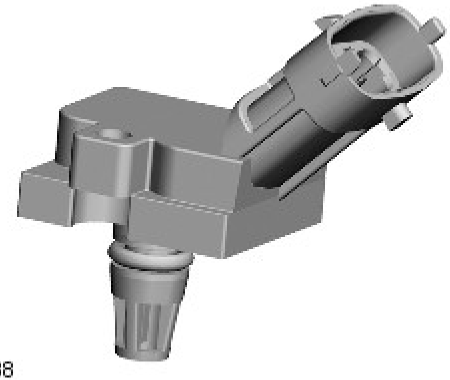
The Manifold Absolute Pressure (MAP) sensor allows the ECM to calculate the load on the engine, which is used in the calculation of fuel injection time.

The MAP sensor is installed in the air inlet of the supercharger. The sensor is secured with a single screw and sealed with an O-ring. A three pin electrical connector provides the interface with the engine harness.

If the MAP sensor fails, the ECM adopts a default value of 1 bar (14.5 lbf/in.2).

With a failed MAP sensor, the engine will suffer from poor starting, rough running and poor drive ability.

MANIFOLD ABSOLUTE PRESSURE AND TEMPERATURE (MAPT) SENSOR



E116088

The Manifold Absolute Pressure and Temperature (MAPT) sensor allows the Engine Control Module (ECM) to calculate the charge air density immediately before it enters the cylinders. This is used to adjust the ignition timing relative to the boost pressure, and to monitor the performance of the charge air coolers.

The MAPT sensor is installed in the rear of the left intake manifold. The sensor is secured with a single screw and sealed with an O-ring. A four pin electrical connector provides the interface with the engine harness.

MASS AIR FLOW AND TEMPERATURE (MAFT) SENSOR

E148513

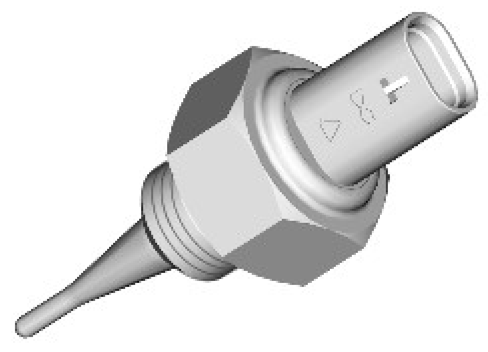
The Mass Air Flow and Temperature (MAFT) sensors allow the Engine Control Module (ECM) to measure the mass flow and the temperature of the air flow into the engine. The mass air flow is measured with a hot film element in the sensor. The temperature of the air flow is measured with a Negative Temperature Coefficient (NTC) thermistor in the sensor. The mass air flow is used to determine the fuel quantity to be injected in order to maintain the stoichiometric air:fuel mixture required for correct operation of the engine and the catalytic converters.

Identical MAFT sensors are attached to each of the air cleaner outlets. On each MAFT sensor, a five pin electrical connector provides the interface with the engine harness.

If the hot film element signal fails the ECM invokes a software backup strategy to calculate the mass air flow from other inputs. Closed loop fuel control, closed loop idle speed control and evaporative emissions control are discontinued. The engine will suffer from poor starting, poor throttle response and, if the failure occurs while driving, the engine speed may dip before recovering.

If the NTC thermistor signal fails the ECM adopts a default value of 25 °C (77 °F) for the intake air temperature.

CHARGE AIR TEMPERATURE SENSOR



E149162

The charge air temperature sensor is installed in the supercharger top cover. A two pin electrical connector provides the interface between the sensor and the engine harness. The sensor contains an Negative Temperature Coefficient (NTC) thermistor with supply and return connections to the Engine Control Module (ECM).

The ECM supplies the charge air temperature sensor with a 5 Volt (V) reference voltage and translates the return voltage into a temperature. The ECM uses the input:

- To monitor operation of the charge air coolant pump.
- For air mass calculations used in control of the supercharger bypass valve, as part of the charge air strategy that co-ordinates operation of the electric throttle and the bypass valve, and predicts the air mass delivered to the cylinders.

NOTE:

The charge air temperature sensor is introduced together with the electric actuator for the supercharger bypass valve (in place of the vacuum actuator), which enables more accurate control of the bypass air flow.

If the charge air temperature sensor fails, the ECM substitutes the input with a modeled temperature. Failure of the sensor is unlikely to be noticeable to the driver.

THROTTLE POSITION SENSOR (TPS)

The Throttle Position Sensor (TPS) allows the ECM to determine the position and angular rate of change of the throttle blade.

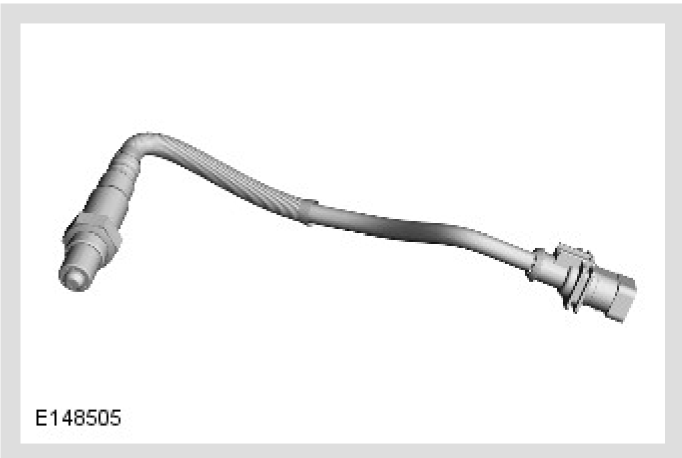
There are two TPS located in the electric throttle. See below for details of the electric throttle.

If a TPS fails, the ECM:

- Adopts a limp home mode where engine speed is limited to a maximum of approximately 2000 Revolutions Per Minute (RPM).
- Discontinues evaporative emissions control.
- Discontinues closed loop control of engine idle speed.

With a failed TPS, the engine will suffer from poor running and throttle response.

HEATED OXYGEN SENSOR (HO2S)



The Heated Oxygen Sensor (HO2S) allow the ECM to measure the oxygen content of the exhaust gases, for closed loop control of the fuel and air mixture and for catalytic converter monitoring.

A pre-catalyst HO2S is installed in the outlet of each exhaust manifold, which enables independent control of the fuel and air mixture for each cylinder bank. A mid catalyst HO2S is installed in a central position on the side of each catalytic converter and a post catalyst HO2S is installed in each catalytic converter outlet pipe in the exhaust system; these enable the operation of the catalytic converter to be optimized and monitored.

HO2S need to operate at high temperatures in order to function correctly. To achieve the high temperatures required, the sensors are fitted with heater elements that are

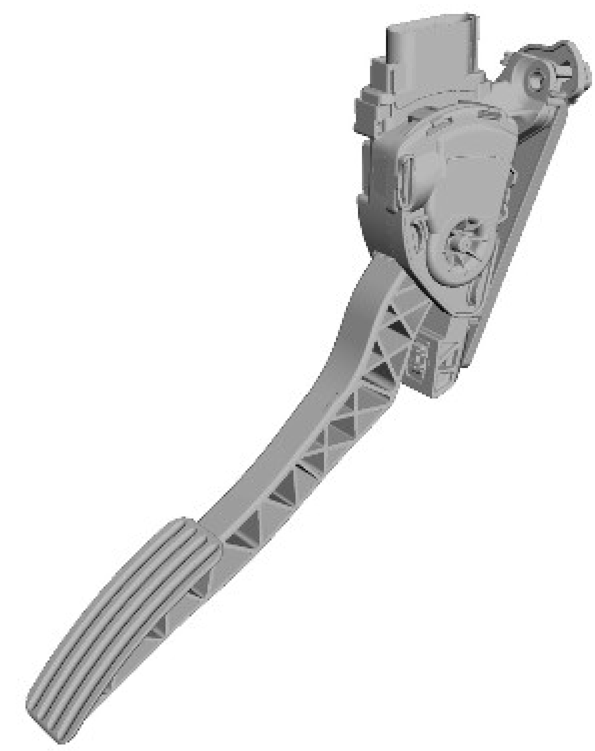
controlled by a Pulse Width Modulation (PWM) signal from the ECM. The heater elements are operated after each engine start, once it has been calculated that there is no moisture in the exhaust, and also during low load conditions when the temperature of the exhaust gases is insufficient to maintain the required sensor temperature. The time period for operation of the heater elements is determined by the temperature of the post catalyst HO2S. The PWM duty cycle is carefully controlled to prevent thermal shock to cold sensors. A non-functioning heater delays the sensor’s readiness for closed loop control and increases emissions.
For additional information, refer to: Electronic Engine Controls - DTC: Heated Oxygen Sensors (HO2S) (303-14 Electronic Engine Controls - V8 S/C 5.0L Petrol, Diagnosis and Testing).

The pre catalyst HO2S produce a constant voltage, with a variable current that is proportional to the lambda ratio. The mid and post catalyst HO2S produce an output voltage dependant on the ratio of the exhaust gas oxygen to the ambient oxygen.

Diagnosis of electrical faults only begin once the sensors have reached operational temperature and become active. This is achieved by checking the signal against maximum and minimum threshold, for open and short circuit conditions.

If HO2S fails, the engine performance may suffer, depending on which sensor it is, however the Malfunction Indicator Lamp (MIL) will always be displayed if the HO2S failed.

ACCELERATOR PEDAL POSITION (APP) SENSOR


E141165

The Accelerator Pedal Position (APP) sensor allows the ECM to determine the driver requests for vehicle speed, acceleration and deceleration. The ECM uses this information, together with information from the Anti-lock Brake System (ABS) control module and the Transmission Control Module (TCM), to determine the setting of the electric throttle.

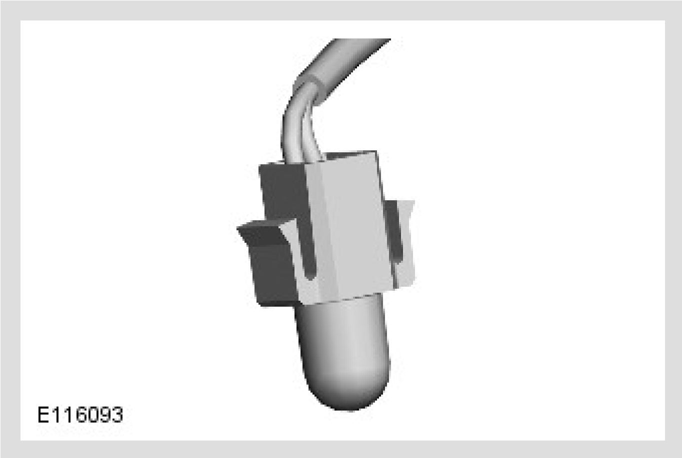
Three screws attach the APP sensor and integrated accelerator pedal to a bracket on the lower instrument panel. A six pin electrical connector provides the interface with the vehicle harness.

The APP sensor is a twin track potentiometer. Each track receives an independent power supply from the ECM and returns an independent analog signal to the ECM. Both signals contain the same positional information, but the signal from track 2 is half the voltage of the signal from track 1 at all positions.

If both signals have a fault, the ECM adopts a limp home mode, which limits the engine speed to 2000 Revolutions Per Minute (RPM) maximum.

The ECM constantly checks the range and plausibility of the two signals and stores a fault code if it detects a fault.

AMBIENT AIR TEMPERATURE (AAT) SENSOR



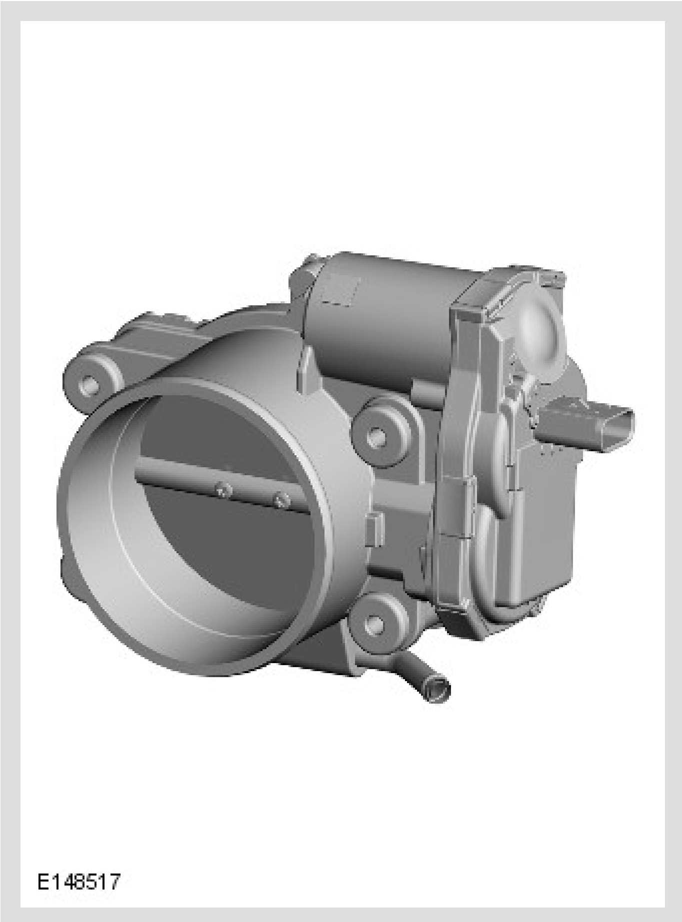
The Ambient Air Temperature (AAT) sensor is a Negative Temperature Coefficient (NTC) thermistor that allows the Engine Control Module (ECM) to monitor the temperature of the air around the vehicle. The ECM uses the AAT input for a number of functions, including engine cooling fan control. The ECM also transmits the ambient temperature on the HS (High Speed) CAN (Controller Area Network) bus for use by other control modules.

The AAT sensor is installed in the left exterior mirror, with the bulb of the sensor positioned over a hole in the bottom of the mirror casing.

The ECM supplies the sensor with a 5 Volt (V) reference voltage and a ground, and translates the return signal voltage into a temperature.

If there is a fault with the AAT sensor, the ECM calculates the AAT from the temperature inputs of the MAFT sensors. If the AAT sensor and the temperature inputs of the MAFT sensors are all faulty, the ECM adopts a default ambient temperature of 25 °C (77 °F).

ELECTRIC THROTTLE



The Engine Control Module (ECM) uses the electric throttle to regulate engine torque.

The electric throttle is installed between the clean air duct and the supercharger.
For additional information, refer to: Intake Air Distribution and Filtering (303-12 Intake Air Distribution and Filtering - V8 S/C 5.0L Petrol, Description and Operation).

The throttle plate is operated by an electric Direct Current (DC) motor integrated into the throttle body. The ECM uses a Pulse Width Modulation (PWM) signal to control the DC motor. The ECM compares the Accelerator Pedal Position (APP) sensor inputs against an electronic map to determine the required position of the throttle plate. The ECM and electric throttle are also required to:

- Monitor requests for cruise control operation.
- Automatically operate the electric throttle for accurate cruise control.

- Perform all dynamic stability control engine interventions.
- Monitor and carry out maximum engine speed and road speed cut outs.
- Provide different engine maps for the ride and handling optimization system.

A software strategy within the ECM calibrates the position of the throttle plate at the beginning of each ignition cycle. When the ignition is turned on, the ECM performs a self test and calibration routine by fully closing the throttle plate and then opening it again. This tests the default position springs and allows the ECM to learn the fully closed position.

OPERATION

ENGINE CONTROL MODULE (ECM) RELAY

The Engine Control Module (ECM) relay is used to initiate the power up and power down routines within the ECM. The ECM relay is installed in the Engine Junction Box (EJB).

When the ignition is turned on, battery voltage is applied to the ignition sense input from the Central Junction Box (CJB). The ECM then starts its power up routines and energizes the ECM relay.

When the ignition is turned off, the ECM maintains its powered up state while it conducts the power down routines. This can be for:

- Up to 20 minutes in extreme cases, when the Diagnostic Monitoring Tank Leakage (DMTL) system is running (North American Specification (NAS) markets).
- Up to five minutes when cooling fans are required.

On completion of the power down routines the ECM de-energizes the ECM relay.

ENGINE CONTROL MODULE (ECM) ADAPTIONS

The Engine Control Module (ECM) has the ability to adapt the input values it uses to control certain outputs. This capability maintains engine refinement and ensures the engine emissions remain within the legislated limits. The components which have adaptions associated with them are:

- The Accelerator Pedal Position (APP) sensor.
- The Heated Oxygen Sensor (HO2S) (6 off).
- The Mass Air Flow and Temperature (MAFT) sensor (2 off).
- The Crankshaft Position (CKP) sensor.
- Electric throttle.

- Knock sensor (2 off).

HEATED OXYGEN SENSOR (HO2S) AND MASS AIR FLOW AND TEMPERATURE (MAFT) SENSOR

There are several adaptive maps associated with the fueling strategy. Within the fueling strategy the Engine Control Module (ECM) calculates short-term adaptions and long term adaptions. The ECM will monitor the deterioration of the HO2Ss over a period of time. It will also monitor the current correction associated with the sensors.

The ECM will store a fault code in circumstances where an adaption is forced to exceed its operating parameters. Simultaneously, the ECM will record the engine speed, engine load and intake air temperature.

CRANKSHAFT POSITION (CKP) SENSOR

The characteristics of the signal supplied by the Crankshaft Position (CKP) sensor are learned by the Engine Control Module (ECM). This enables the ECM to set an adaption and support the engine misfire detection function. Due to a small variation between different drive plates and different CKP sensors, the adaption must be reset if either component is renewed, or removed and refitted. It is also necessary to reset the drive plate adaption if the ECM is renewed or replaced. The ECM supports four drive plate adaptions for the CKP sensor. Each adaption relates to a specific engine speed range. The engine speed ranges are detailed in the table below:

ADAPTION	ENGINE SPEED - REVOLUTION PER MINUTE (RPM)
1	1800 - 3000
2	3001 - 3800
3	3801 - 4600
4	4601 - 5400

MISFIRE DETECTION

Legislation requires that the ECM must be able to detect the presence of an engine misfire. It must be able to detect misfires at two separate levels. The first level is an amount of misfire that could lead to the legislated emissions limit being exceeded by a given amount. The second level is a misfire rate that causes degradation in catalytic converter efficiency.

The ECM monitors the number of misfire occurrences within two engine revolution ranges. If the ECM determines a misfire failure within either of these two ranges, over two consecutive journeys, it will record a fault code and details of the engine speed, engine load and engine coolant temperature. In addition, if the second level of misfire occurs, on any trip, the ECM flashes the MIL (Malfunction Indicator Lamp) while the fault is occurring.

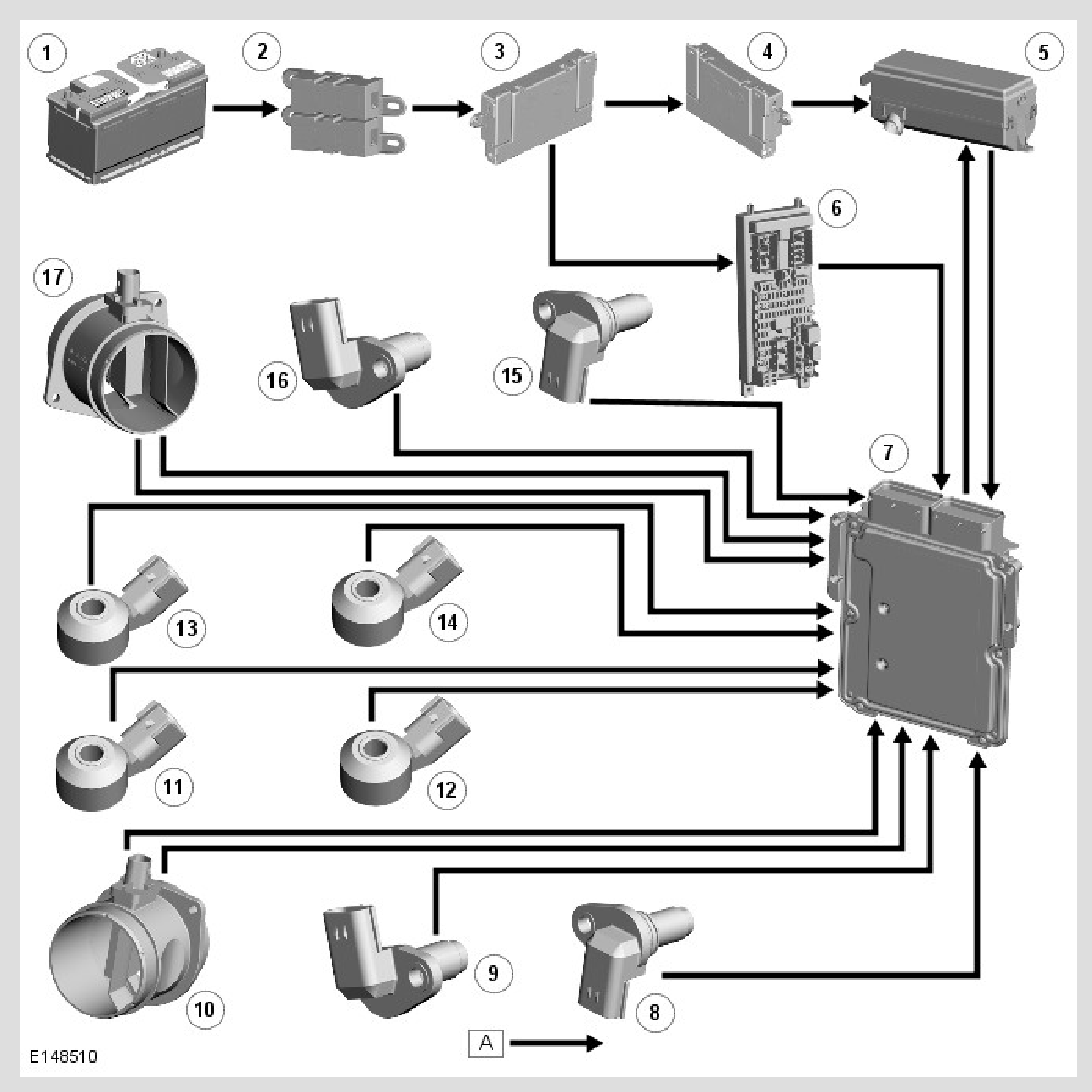
The signal from the CKP sensor indicates how fast the poles on the drive plate are passing the sensor tip. A sine wave is generated each time a pole passes the sensor tip. The ECM can detect variations in drive plate speed by monitoring the sine wave signal supplied by the crankshaft position sensor. By assessing this signal, the ECM can detect the presence of an engine misfire. The ECM will evaluate the signal against a number of factors and will decide whether to record the occurrence or ignore it. The ECM can assign a misfire judgment to an individual cylinder, which can be viewed on Land Rover approved diagnostic equipment.

DIAGNOSTICS

The Engine Control Module (ECM) stores each fault as a Diagnostic Trouble Code (DTC). The DTC and associated environmental and freeze frame data can be read using Land Rover approved diagnostic equipment, which can also read real time data from each sensor, the adaption values currently being employed and the current fueling, ignition and idle speed settings.

CONTROL DIAGRAM

Diagram 1 of 2

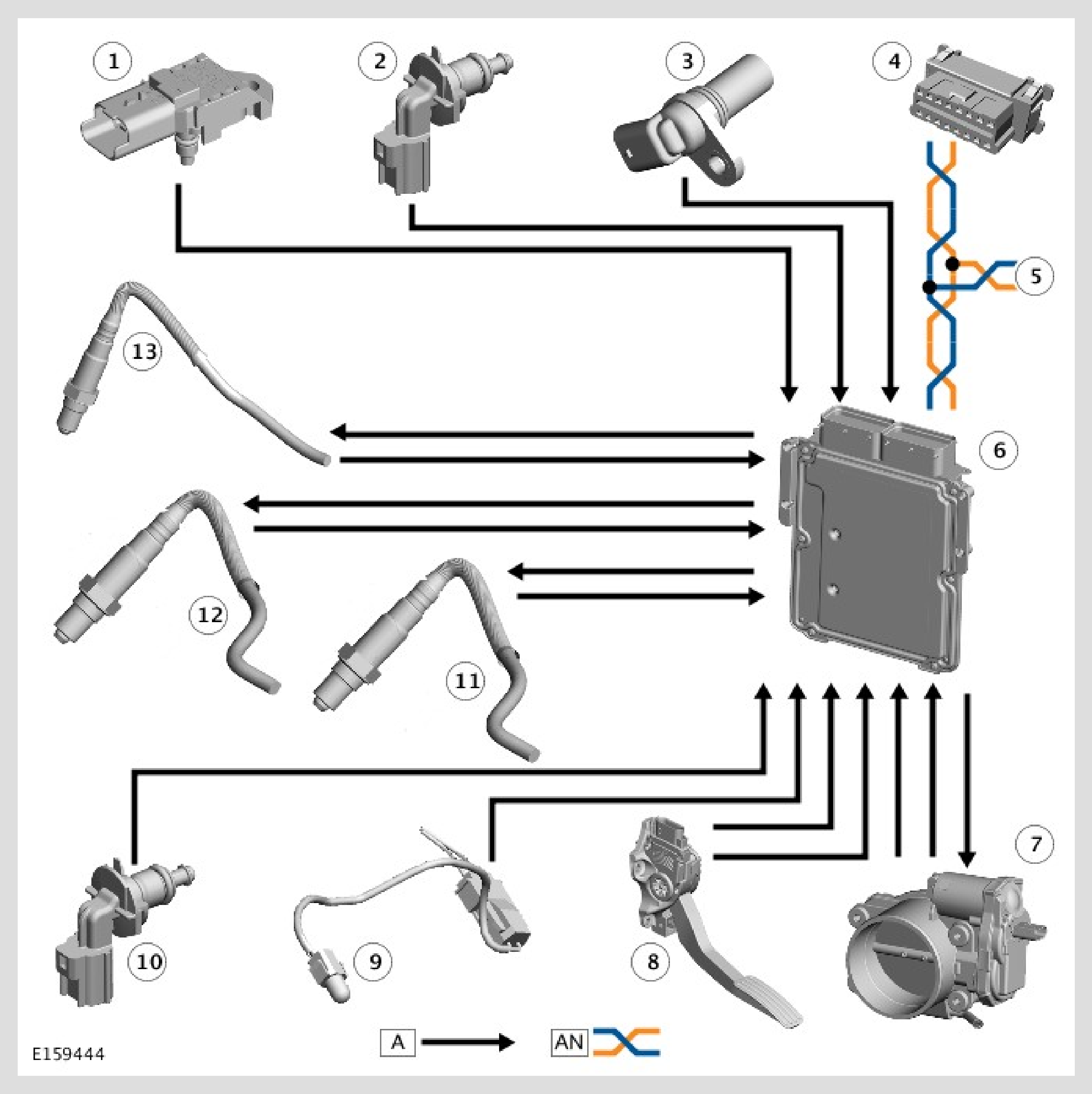


A = HARDWIRED.

ITEM	DESCRIPTION
1	Battery
2	Battery Junction Box 2 (BJB2)
3	Battery Junction Box (BJB)
4	Auxiliary Junction Box (AJB)
5	Engine Junction Box (EJB)
6	Central Junction Box (CJB)
7	Engine Control Module (ECM)
8	Left intake Camshaft Position (CMP) sensor
9	Left exhaust Camshaft Position (CMP) sensor
10	Left Mass Air Flow and Temperature (MAFT) sensor

11	Left front knock sensor
12	Left rear knock sensor
13	Right rear knock sensor
14	Right front knock sensor
15	Right intake Camshaft Position (CMP) sensor
16	Right exhaust Camshaft Position (CMP) sensor
17	Right Mass Air Flow and Temperature (MAFT) sensor

Diagram 2 of 2



A = HARDWIRED; AN = HIGH SPEED (HS) CONTROLLER AREA NETWORK (CAN) POWERTRAIN BUS.

ITEM	DESCRIPTION
1	Manifold Absolute Pressure (MAP) sensor
2	Engine Coolant Temperature (ECT) sensor 2
3	Crankshaft Position (CKP) sensor
4	Diagnostic connector

5	High Speed (HS) Controller Area Network (CAN) powertrain bus connection to other system control modules
6	Engine Control Module (ECM)
7	Electric throttle
8	Accelerator Pedal Position (APP) sensor
9	Ambient Air Temperature (AAT) sensor
10	Engine Coolant Temperature (ECT) sensor 1
11	Heated Oxygen Sensor (HO2S) - Mid catalyst (2 off)
12	Heated Oxygen Sensor (HO2S) - Post catalyst (2 off)
13	Heated Oxygen Sensor (HO2S) - Pre catalyst (2 off)
14	Manifold Absolute Pressure and Temperature (MAPT) sensor
15	Charge air temperature sensor